

Azure Service Bus – Correlation Filters

1. What is a Correlation Filter?

A **Correlation Filter** in Azure Service Bus is used to filter messages based on **equality conditions** on message properties.

In simple words:

A subscription receives a message only when the message properties match the conditions defined in the correlation filter.

Example

Suppose we have a topic called:

OrdersTopic

And a subscription:

S1

We create a correlation filter:

City = Mumbai

Now:

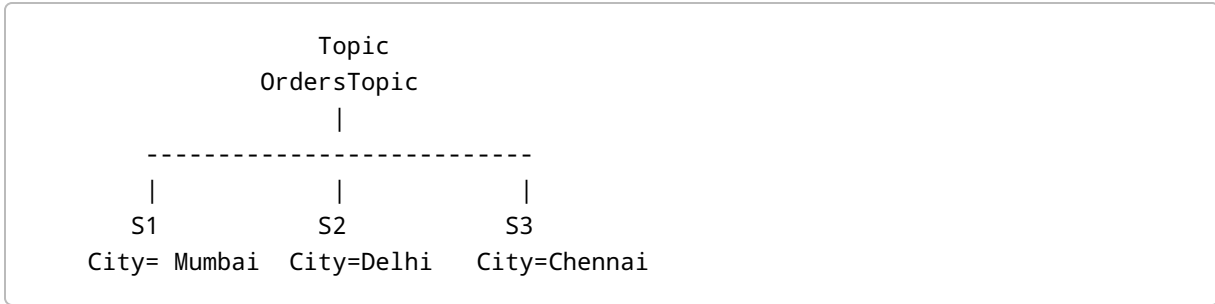
- Message with City = Mumbai → ✓ Delivered to S1
- Message with City = Bangalore → ✗ Not delivered to S1
- Message without City → ✗ Not delivered to S1

2. Why Do We Use Correlation Filters?

A topic can have multiple subscriptions.

Each subscription may be interested in different types of messages.

For example:



Each subscription receives only the messages that satisfy its filter.

This allows us to implement **message routing**.

3. Correlation Filter Example

Suppose we create subscription **S1** with this filter:

```
City = Mumbai
```

When a message is sent to the topic:

```
Message
-----
OrderId = 101
Product = Laptop
City = Mumbai
```

The filter checks:

```
City == Mumbai
```

Result:

```
true
```

Therefore:

```
Message → S1
```

But if the message is:

```
OrderId = 102  
Product = Mobile  
City = Bangalore
```

The filter checks:

```
City == Mumbai
```

Result:

```
false
```

Therefore:

```
Message → S1 ❌
```

4. Important Concept – Application Properties

The correlation filter can check **application/user-defined message properties**.

For example:

```
City = Mumbai
```

Here:

- `City` → Property name
- `Mumbai` → Property value

These properties are attached to the Service Bus message.

Example:

```
Message Body:  
Order Created
```

```
Application Properties:  
City = Mumbai
```

The correlation filter checks the application property.

5. Creating a Subscription with a Correlation Filter

We first create a Service Bus Administration Client.

```
var adminClient =  
    new ServiceBusAdministrationClient(connectionString);
```

The administration client is used to perform management operations such as:

- Create a topic
 - Create a subscription
 - Create rules/filters
 - Delete subscriptions
 - Manage Service Bus entities
-

6. Create Subscription Options

We create subscription details using:

```
var s1Details =  
    new CreateSubscriptionOptions(topicName, "S1");
```

Here:

```
topicName → Topic where subscription should be created  
  
"S1" → Name of the subscription
```

7. Create Correlation Filter

Now we create a correlation filter.

```
var s1Rule =
    new CreateRuleOptions(
        "S1Rule",
        new CorrelationRuleFilter
        {
            ApplicationProperties =
            {
                { "City", "Mumbai" }
            }
        }
    );
```

The important part is:

```
{ "City", "Mumbai" }
```

This means:

```
Application Property:
City = Mumbai
```

Therefore, the subscription accepts messages where:

```
City == Mumbai
```

8. Create the Subscription

Finally, we create the subscription:

```
await adminClient.CreateSubscriptionAsync(
    s1Details,
    s1Rule);
```

Now Azure Service Bus creates:

```
Topic
├── S1
│   └──
```

```
└─ S1Rule
   └─ Correlation Filter
       City = Mumbai
```

9. How Message Filtering Works

Suppose we send this message:

```
Message Body:
Order Created
```

```
Application Properties:
City = Mumbai
```

The filter checks:

```
Message City = Mumbai
Filter City  = Mumbai
```

Both are equal.

Therefore:

```
Message → S1 
```

If we send:

```
Message Body:
Order Created
```

```
Application Properties:
City = Bangalore
```

The filter checks:

```
Message City = Bangalore  
Filter City = Mumbai
```

They are not equal.

Therefore:

```
Message → S1 ❌
```

10. What Happens If We Don't Provide the Property?

Suppose the filter is:

```
City = Mumbai
```

But we send a message without the `City` property.

Example:

```
Message Body:  
Order Created  
  
Application Properties:  
No City property
```

The message does not satisfy the filter.

Therefore:

```
Message → S1 ❌
```

11. Azure Portal Verification

After creating the subscription, we can open:

```
Azure Portal
  ↓
Service Bus Namespace
  ↓
Topic
  ↓
Subscriptions
```

We should see:

```
S1
```

Opening the subscription shows its rule/filter.

Example:

```
Rule Name:
S1Rule

Filter Type:
Correlation Filter

Application Property:
City = Mumbai
```

12. Testing the Correlation Filter

We can use **Service Bus Explorer** to test the filter.

Test 1 - Without City Property

Send:

```
Message:
Hello Service Bus
```

Do not add:


```
City = Mumbai
```

Expected result:

```
S1 Message Count = 0
```

The message is not delivered to S1 because it does not satisfy the filter.

Test 2 - City = Mumbai

Send:

```
Message:  
Hello Service Bus
```

Add the custom application property:

```
Key   = City  
Value = Mumbai
```

Now the filter checks:

```
City == Mumbai
```

Condition is true.

Expected result:

```
S1 Message Count = 1
```

The message is delivered to S1.

Test 3 - City = Bangalore

Send:

```
Message:
Hello Service Bus
```

Add:

```
Key   = City
Value = Bangalore
```

Filter:

```
City = Mumbai
```

Condition:

```
Bangalore == Mumbai
```

Result:

```
false
```

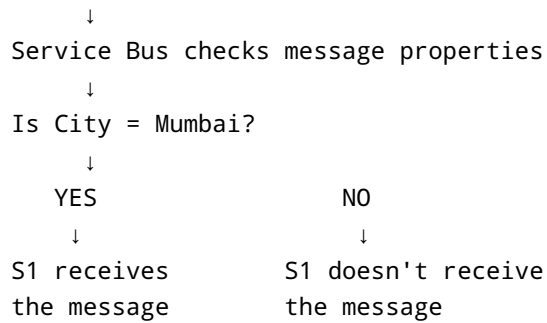
Therefore:

```
S1 does not receive the message.
```

13. Complete Flow

The complete flow can be remembered like this:

```
Create Topic
  ↓
Create Subscription S1
  ↓
Create Correlation Filter
  ↓
City = Mumbai
  ↓
Send Message to Topic
```



14. Correlation Filter vs SQL Filter

Azure Service Bus supports different types of filters.

The two important ones are:

Correlation Filter

Used for simple equality-based matching.

Example:

```
City = Mumbai
```

It is useful when the filtering requirement is straightforward.

SQL Filter

Used when we need more complex conditions.

Example:

```
City = 'Mumbai' AND OrderAmount > 10000
```

So remember:

```
Correlation Filter → Simple equality matching
```

```
SQL Filter → Complex filtering conditions
```

15. Real-Time Example

Imagine an e-commerce application.

We have a topic:

```
OrderTopic
```

And subscriptions:

```
MumbaiOrders  
BangaloreOrders  
DelhiOrders
```

Filters:

```
MumbaiOrders    → City = Mumbai  
BangaloreOrders → City = Bangalore  
DelhiOrders     → City = Delhi
```

Suppose an order message contains:

```
OrderId = 1001  
Customer = Puneeth  
City = Mumbai  
Amount = 5000
```

Azure Service Bus checks the subscriptions.

```
MumbaiOrders  
City = Mumbai  
    ↓  
    MATCH ✓  
  
BangaloreOrders  
City = Bangalore  
    ↓  
    NO MATCH ✗  
  
DelhiOrders  
City = Delhi
```

↓
NO MATCH ❌

Therefore, only:

MumbaiOrders

receives the message.

16. Interview Questions

Q1. What is a correlation filter in Azure Service Bus?

A correlation filter is used to filter messages based on equality conditions on message properties.

Example:

City = Mumbai

Only messages having the application property `City` with value `Mumbai` will be delivered to that subscription.

Q2. Where is a correlation filter applied?

A filter is applied to a **subscription** of a Service Bus topic.

Topic
↓
Subscription
↓
Filter

Q3. What happens if a message doesn't satisfy the filter?

The message is not delivered to that subscription.

It may still be delivered to other subscriptions if their filters match the message.

Q4. Can multiple subscriptions have different filters?

Yes.

For example:

```
S1 → City = Mumbai  
S2 → City = Bangalore  
S3 → City = Delhi
```

The same topic can have multiple subscriptions with different filters.

Q5. What is the difference between a correlation filter and a SQL filter?

Correlation Filter:

```
City = Mumbai
```

Used for simple equality matching.

SQL Filter:

```
City = 'Mumbai' AND OrderAmount > 10000
```

Used for more complex filtering conditions.

Q6. What are application properties?

Application properties are custom key-value properties attached to a Service Bus message.

Example:

```
City = Mumbai  
Priority = High  
CustomerType = Premium
```

These properties can be used by filters to determine whether a message should be delivered to a subscription.

17. Key Points to Remember

- A **correlation filter** performs equality-based filtering.
- Filters are associated with **topic subscriptions**.
- Filtering can be done using **application properties**.
- Example:

```
City = Mumbai
```

- Only messages matching the condition are delivered to that subscription.
- Different subscriptions can have different filters.
- Correlation filters are suitable for simple filtering.
- SQL filters are suitable for more complex conditions.
- If a message doesn't match a subscription's filter, that subscription does not receive the message.
- Service Bus Explorer can be used to test message filtering.

One-Line Interview Answer

A correlation filter in Azure Service Bus is used to route messages to a topic subscription based on equality matching of message properties, such as

City = Mumbai.